THE UNIVERSITY OF MELBOURNE

Student Number

Semester 1 Assessment, 2019

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Writing time: 3 hours

Reading time: 15 minutes

This is NOT an open book exam

This paper consists of 14 pages (including this page)

Authorised Materials

- Mobile phones, smart watches and internet or communication devices are forbidden.
- Calculators, tablet devices or computers must not be used.
- No handwritten or print materials may be brought into the exam venue.

Instructions to Students

- You must NOT remove this question paper at the conclusion of the examination.
- No calculators or notes of any kind may be brought into the examination.
- Two pages of formulae as well as a standard normal distribution table are provided.
- You should attempt all questions. Marks for individual questions are shown.
- **All solutions should be written in this exam booklet.**
- Blank scripts books are provided for rough working only and will not be graded.
- You may use the blank pages in this booklet if you require more space. Please clearly indicate in your solution if you do so. Otherwise, it will not be graded.
- Unless otherwise stated all solutions and clear working should be shown.
- There are 10 questions with marks as shown. The total number of marks available is 100.

Instructions to Invigilators

- Students must NOT remove this question paper at the conclusion of the examination.

This paper may be held in the Baillieu Library

Blank page (ignored in page numbering)

Short-answer questions: Put your answer in the box provided. Full marks will be given for a correct answer placed in the box. Part marks will be given for incorrect answers if working is shown.

Question 1 (27 marks)

(a) Tom owns 6 different mathematics books and 4 different computer science books and wish to fill 5 positions on a shelf. If the first 3 positions are to be occupied by mathematics books and the last 2 by computer science books, in how many ways can this be done? (A numerical answer is required for this question.)

Answer:

(b) Find the equation of the tangent line of $f(x) = (1-x)^2$ about $x = 3$

Answer:

(c) Evaluate the following quantity

$$\sin\left(-\frac{\pi}{4}\right) + \cos\left(\frac{23\pi}{3}\right)$$

Answer:

Blank page (ignored in page numbering)

(d) Let $f(x) = \cos(x)\log(x)$. Find $f'(x)$.

Answer:

(e) Let $g(x) = \dfrac{1 + \frac{1}{x}}{e^x}$. Find $g'(x)$.

Answer:

(f) Let $h(x) = \log(\sin(5x))$. Find $h'(x)$.

Answer:

Blank page (ignored in page numbering)

(g) Evaluate the following:

$$\int \frac{x^4+4}{x}\,dx$$

Answer:

(h) Evaluate the following:

$$\int_0^3 4e^x - \sin(\pi x)\,dx$$

Answer:

(i) Solve the following initial value problem

$$F(x) = \int \frac{2}{\sqrt{x}}\,dx \quad \text{if } F(9) = 0$$

Answer:

Blank page (ignored in page numbering)

Full-solution problems: Complete solutions are required for the remainder of this examination. Be sure to justify your answers and to show all your work. In addition to the accuracy of your answer and methods, you will be assessed on the justification of your solutions. If you require more space, you may use the reverse side of this paper. Please clearly indicate in your solution if you do so. Otherwise, it will not be graded.

Question 2 (7 marks) Find all solutions of the trigonometric equation

$$2\sin(3x) = -\sqrt{3}$$

Blank page (ignored in page numbering)

Question 3 (7 marks)

(a) Evaluate the following limit if it exists. Otherwise, write DNE and justify why the limit does not exist.

$$\lim_{x \to 3} \frac{x^3 - 2x^2 - 6x + 9}{x - 3}$$

(b) Use the limit definition of the derivative to find $f'(x)$ given $f(x) = \frac{1}{x^2}$ for $x \neq 0$. No marks will be awarded for using any other method.

Blank page (ignored in page numbering)

Question 4 (10 marks) Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = -x^2 + 2x + 8$.

Hint: You may use, without proof, that the function can also be written as

$$f(x) = -x^2 + 2x + 8 = (4 - x)(x + 2) = -(x - 1)^2 + 9$$

(a) What is the domain, codomain, and range of f?

(b) Find the largest value q such that the function $f : (-\infty, q] \to \mathbb{R}$ given by

$$f(x) = -x^2 + 2x + 8$$

is injective.

(c) Find the interval C such that $f : (-\infty, q] \to C$ given by

$$f(x) = -x^2 + 2x + 8$$

is surjective.

Blank page (ignored in page numbering)

(d) Based on your choice of q and C, explain why the function $f : (-\infty, q] \to C$ given by $f(x) = -x^2 + 2x + 8$ is invertible.

(e) Define g to be the inverse of f. State the domain and codomain of g as well as provide the algebraic equation for $g(x)$.

Blank page (ignored in page numbering)

Question 5 (11 marks) Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \frac{2}{1+x^2} - 1$.
You may use without proof that

$$f(x) = \frac{2}{1+x^2} - 1 = \frac{1-x^2}{1+x^2}$$

(a) Find all x-intercepts and the y-intercept of the function.

(b) Find the intervals where $f(x)$ is positive and intervals where $f(x)$ is negative

(c) Evaluate and simplify $f'(x)$

(d) Find the points in the domain where $f'(x) = 0$.

(e) Find the intervals where $f(x)$ is increasing and intervals where $f(x)$ is decreasing

(f) Find and classify the location of any local extrema of the function.

Blank page (ignored in page numbering)

Question 6 (11 marks) A particle travels along the path described by the curve $y = 2\sqrt{x+6}$.

(a) Use graph transformations to sketch a graph of the path. Clearly label any intercepts.

(b) The distance between any two points (a, b) and (p, q) in the Cartesian plane is given by

$$d = \sqrt{(p-a)^2 + (q-b)^2}$$

Find the distance between the particle at a point (x, y) along path and the origin $(0, 0)$ as a function in terms of x only.

(c) Using this information, find the coordinates of the point along the path that is closest to the origin and justify your answer.

Blank page (ignored in page numbering)

Question 7 (10 marks) Consider two functions given by

$$y = x^2 + x \qquad \text{and} \qquad y = 3 - x$$

(a) Find the coordinates where the two functions intersect.

(b) Find the **physical** area enclosed by the two functions between $x = -3$ and $x = 3$.

Blank page (ignored in page numbering)

Question 8 (5 marks) Consider the discrete probability distribution for the random variable X shown in the table below:

x	-1	0	1	2
$P(X = x)$	0.2	0.1	0.4	0.3

(a) Explain why the data above forms a valid discrete probability distribution.

(b) Find $\mathbb{E}(X)$.

(c) Given that the outcome is strictly positive, find the probability that $X = 2$.

Blank page (ignored in page numbering)

Question 9 (7 marks) The time taken for a person to walk from Agar theatre to Peter Hall building normally distributed with mean 13 minutes and an unknown variance $V\text{min}^2$.

(a) The quickest 9% of people can complete the walk in 10.32mins. Compute a value of V using this information.

(b) A poll was conducted and the results estimated that the variance for this walk is 16mins^2. Use this information to find the percentage of people who can complete the walk in between 11 minutes and 19 minutes.

Blank page (ignored in page numbering)

Question 10 (5 marks) *Careful reasoning is required for this question. Even the right answer with insufficient justification will not score well in this question.*

The question refers to the set of functions $f : \{e, p, s, t\} \to \{5, 6, 7, 8\}$.

(a) Write down a function f with this domain and codomain.

(b) Find the number of different functions with this domain and codomain.

(c) Find the number of different *bijective* functions with this domain and codomain.

End of exam questions—Total Available Marks = 100
Turn the page for appended material

Blank page (ignored in page numbering)

Formula Sheet

- **Combinatorics:**

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \ldots \cdot 3 \cdot 2 \cdot 1$$
$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

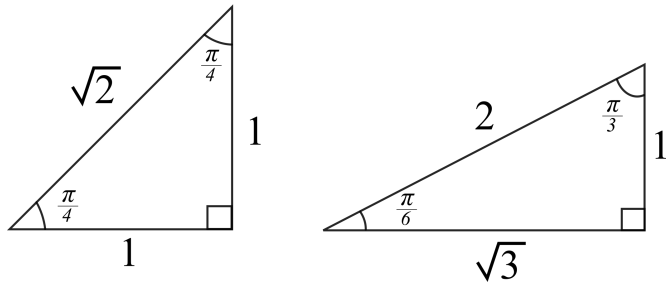
- **Functions:**
  - Lines

$$y = mx + c \qquad\qquad y - b = m(x-a)$$

  - Quadratics

$$y = ax^2 + bx + c = a(x-r)(x-s) = a(x-h)^2 + k$$
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Special Triangles:**



- **Trigonometric identities:**

$$\sin(x + 2\pi) = \sin(x) \qquad \cos(x + 2\pi) = \cos(x)$$
$$\sin(\pi - x) = \sin(x) \qquad \sin(-x) = -\sin(x)$$
$$\cos(\pi - x) = -\cos(x) \qquad \cos(-x) = \cos(x)$$

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$$
$$\sin(2x) = 2\sin(x)\cos(x) \qquad \cos(2x) = \cos^2(x) - \sin^2(x)$$
$$\sin^2(x) = \frac{1 - \cos(2x)}{2} \qquad \cos^2(x) = \frac{1 + \cos(2x)}{2}$$

- **Exponential and logarithms:**

$$a^0 = 1 \qquad a^1 = a$$
$$a^{-1} = \frac{1}{a} \qquad a^{\frac{1}{2}} = \sqrt{a}$$
$$a^x \cdot a^y = a^{x+y} \qquad \frac{a^x}{a^y} = a^{x-y}$$
$$(a^x)^y = a^{x \cdot y} \qquad \sqrt[y]{a^x} = a^{\frac{x}{y}}$$
$$a^x \cdot b^x = (ab)^x \qquad \frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

$$\log_e(x) = \ln(x) \qquad \log_a(1) = 0$$
$$\log_a(a) = 1 \qquad \log_a(p^x) = x\log_a(p)$$
$$\log_a(pq) = \log_a(p) + \log_a(q) \qquad \log_a\left(\frac{p}{q}\right) = \log_a(p) - \log_a(q)$$
$$\log_a(p) = \frac{\log_c(p)}{\log_c(a)}$$

- **Limits:** If we have

$$\lim_{x\to a} f(x) = K \qquad \lim_{x\to a} g(x) = L$$

Then

$$\lim_{x\to a} [f(x) + g(x)] = K + L \qquad \lim_{x\to a} [f(x) - g(x)] = K - L$$

$$\lim_{x\to a} [f(x) \times g(x)] = K \times L \qquad \lim_{x\to a} \left[\frac{f(x)}{g(x)}\right] = \frac{K}{L}$$

$$\lim_{x\to a} [f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n \qquad \lim_{x\to a} [f(x)]^{\frac{1}{n}} = \left[\lim_{x\to a} f(x)\right]^{\frac{1}{n}}$$

$$\lim_{x\to a} cf(x) = cK$$

$$\lim_{x\to a} c = c \qquad \lim_{x\to a} x = a$$

$$\lim_{x\to a} x^n = a^n \qquad \lim_{x\to a} x^{\frac{1}{n}} = a^{\frac{1}{n}}$$

- **Derivatives:**

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{a\to x} \frac{f(a) - f(x)}{a - x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(c\Big) = 0 \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(x^k\Big) = kx^{k-1}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\sqrt{x}\Big) = \frac{1}{2\sqrt{x}} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(e^x\Big) = e^x$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\log(x)\Big) = \frac{1}{x} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\sin(x)\Big) = \cos(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\cos(x)\Big) = -\sin(x) \qquad \frac{\mathrm{d}}{\mathrm{d}x}\Big(\tan(x)\Big) = \frac{1}{\cos^2(x)}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(u(x)\cdot v(x)\Big) = \frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) + u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x} = u'(x)\cdot v(x) + u(x)\cdot v'(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(\frac{u(x)}{v(x)}\Big) = \frac{\frac{\mathrm{d}u}{\mathrm{d}x}\cdot v(x) - u(x)\cdot\frac{\mathrm{d}v}{\mathrm{d}x}}{[v(x)^2]} = \frac{u'(x)\cdot v(x) - u(x)\cdot v'(x)}{[v(x)^2]}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\Big(f(u(x))\Big) = \frac{\mathrm{d}f}{\mathrm{d}u}\frac{\mathrm{d}u}{\mathrm{d}x} = f'(u)\cdot u'(x)$$

The standard normal distribution table for $P(X < x)$

x	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000